

ORIGINAL ARTICLE

***Obelia* (Cnidaria, Hydrozoa, Campanulariidae):
a microphagous, filter-feeding medusa**Ferdinando Boero¹, Cecilia Bucci¹, Anna Maria Rosaria Colucci¹, Cinzia Gravili¹ & Loredana Stabili^{1,2}¹ Dipartimento di Scienze e Tecnologie Biologiche ed Ambientali (Di.S.Te.B.A.), University of Salento, Via Prov.le Lecce-Monteroni, Lecce, Italy² Istituto Ambientale Marino Costiero, Sezione di Taranto, CNR, Taranto, Italy**Keywords**Ecology; feeding strategy; Hydrozoa; Ionian Sea; microphagy; *Obelia*.**Correspondence**

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Abstract

Obelia is used in many zoology textbooks to exemplify the Hydrozoa life cycle. The body plan of *Obelia* medusae, however, is unique among the Hydrozoa. Contrary to all other hydromedusae, *Obelia* medusae do not derive from a medusary nodule, have neither a velum nor a subumbrellar cavity, the umbrella is flat, and swimming is achieved by umbrellar flapping and does not involve jet propulsion. The medusae of most *Obelia* species do not grow well on a diet of *Artemia salina* nauplii, the usual food for laboratory-reared hydrozoans, and their rearing is often difficult. Previous observations reported that *Obelia* medusae might eat phytoplankton, driven to the mouth by movements of both the bell and the tentacles. To evaluate the efficiency of this feeding mechanism, we gave a suspension of genetically modified *Escherichia coli* expressing GFP to starved, newly released medusae of both *O. dichotoma* and *Clytia hemisphaerica* (the other campanulariid genus with medusae). *Obelia* medusae are able to concentrate bacteria in their gastric cavity, whereas *Clytia* medusae are not. Contrary to other Hydromedusae that are macrophagous, *Obelia* is a microphagous and filter-feeding medusa, at least at the onset of its medusan life. As Cnidaria evolved before their current metazoan prey, their ancestral diet was probably microphagous. *Obelia*'s microphagous feeding habits could thus represent an example of the ancestral feeding condition in Cnidaria, but the apomorphic characters of its medusa suggest that its body architecture is not ancestral, and that its microphagy evolved relatively recently.

Problem

Cnidaria are among the oldest Metazoa (Parsons 1979); they are usually ranked as a 'low' animal phylum because of their tissue-level, diploblastic organisation, lacking the true organs characteristic of the 'higher' triploblastic Metazoa. The Hydrozoa is a group of cnidarians that can have both the polyp and the medusa (the Hydroidomedusae, with many cases of medusa suppression), or only the medusa (the Automedusae) (Bouillon & Boero 2000).

The polyp generation of the Hydrozoa plays an important role in shallow marine ecosystems (Boero 1984; Coma *et al.* 1995; Gili & Hughes 1995); hydrozoan colonies are

ephemeral and grow rapidly, and their hydranths are able to catch and consume a great variety of prey. Littoral zooplankton is the main component of hydroids' natural diet (Simkina 1980; Barangé & Gili 1988; Gili *et al.* 1996), although some species feed mainly from the benthos (Christensen 1967). Several studies have reported dinoflagellates, diatoms, tintinnids and bacteria in hydroid gut contents (Simkina 1980; Cornelius & Östman 1987; Barangé & Gili 1988; Hunter 1989).

The medusae of the Hydrozoa belong to the carnivorous gelatinous plankton. When jellyfish and ctenophores occur in large populations, their predation can have substantial effects on plankton food webs, as they are

voracious consumers of both zoo- and ichthyoplankton (Purcell 1997; Purcell & Decker 2005). Hydromedusae, including the young of the large species, feed on small-sized particles (mesozooplankton) such as small crustaceans, fish eggs and larvae, meroplanktonic larvae and microzooplankton including ciliates (Purcell & Arai 2001; Purcell & Sturdevant 2001; Costello & Colin 2002; Parsons & Lalli 2003). In the laboratory rearing of hydromedusae to reconstruct life cycles, food sources are invariably based on crustaceans.

The genus *Obelia* belongs to the family Campanulariidae (Hydrozoa, Leptomedusae), an important and widely distributed family of hydrozoans (Cornelius 1982). *Obelia* is a very popular and widespread medusa, used as an example of hydromedusa in many zoology textbooks (e.g. Barnes *et al.* 1988; Ruppert & Barnes 1994). These medusae, however, are very different from all other medusae (Boero & Sarà 1987; Cornelius 1990; Boero *et al.* 1996). The development in the gonotheca starts with a medusary nodule, but this embryological stage soon loses its characteristics and further development resembles that of hydranths: the marginal tentacles are chordal, like those of hydroids; the manubrium is expanded around the mouth, with a sharp proximal constriction resembling the mouth region of campanulariid hydroids; the umbrella is flat and thin, with no subumbrellar cavity; the umbrellar marginal statocysts are at the tentacles bases and not along the manubrial margin, like those of all other hydromedusae; the velum is absent. Because of these peculiarities, swimming in *Obelia* differs considerably from that of other medusae. Contraction of the umbrella can be undulatory (by undular contraction of the tubular circular muscle) or complete (Boero *et al.* 1996). For these reasons, Boero & Sarà (1987) and Boero *et al.* (1996) suggested that *Obelia* medusae might be the result of saltational evolution leading to medusa re-expression in a lineage with suppressed medusae. Cornelius (1990) did not share this opinion, and Govindarajan *et al.* (2006) discussed it against molecular evidence. *Clytia* is the other campanulariid genus with true medusae. Contrary to the former, however, *Clytia* medusae are similar to those of other families (Boero *et al.* 1996) and grow well on a crustacean diet under laboratory conditions (see Jarms *et al.* 2002 for a review on the cultivation of polyps and medusae).

Kubota (1999) reported for the first time the complete life cycle of the nominal species *O. dichotoma* from Northern Japan. *Obelia* medusae fed on newly hatched *Artemia* nauplii, as described by Kubota (1981). F. Boero and C. Gravili (unpublished data), however, failed in rearing *O. dichotoma* medusae from the Mediterranean Sea; they fed them individually by hand with newly hatched *Artemia* nauplii, even reduced in small pieces, integrating their diet with a suspension based on microalgae.

Boero & Sarà (1987) observed that new-born *Obelia longissima* medusae have no nematocysts on the lips and continuously pulsate, thus creating water currents towards the mouth that enable them to ingest very small food particles. These observations suggested the hypothesis of a microphagous feeding strategy. Such microphagy might exert an impact on the water column by reducing bacterioplankton abundance. Bacteria are an important component of planktonic systems (Cole *et al.* 1988); their abundance in the marine environment, somewhat over-simplified, usually ranges from 10^3 to 10^6 ml⁻¹ (Stolp 1988). In coastal ecosystems, heterotrophic bacterioplankters significantly contribute to the microbial biomass (Fuhrman & Azam 1980); up to 50–60% of the total carbon flow passes through bacterioplankters (Newell & Linley 1984) and several studies have shown their role in the diet of different filter feeders (Orejas *et al.* 2000a, b). The impact of grazing on bacteria may be crucial, especially in oligotrophic marine ecosystems such as the Ionian Sea, where heterotrophic bacteria constitute a major biomass component and the microbial loop represents a biological force driving energy fluxes and ecosystem functioning (Fuhrman *et al.* 1989; Cho & Azam 1990).

In the present study we tested the hypothesis that, contrary to other medusae (namely *Clytia hemisphaerica*), newly released *Obelia* medusae are able to actively concentrate bacteria in their gastric cavity.

Material and Methods

Sampling

Obelia dichotoma (Fig. 1A) and *Clytia hemisphaerica* colonies were collected by scuba diving along the Ionian coasts of Apulia (Le Orte, Otranto Channel, Italy) (40°08'39.8" N, 18°30'23.3" E) at 15–20 m depth in February 2005. They were transported to the laboratory under controlled temperature and placed in a 2-l tank in a temperature-controlled room (14 °C). The salinity of the sterile seawater (0.2 µm pore-filtered) was 37 ppt. Under these experimental conditions, *Obelia dichotoma* and *Clytia hemisphaerica* colonies released medusae 1 day after collection. These were isolated and starved for 1 day to remove any food deriving from the mother colony.

Obelia and *Clytia* medusae were filmed with a video camera connected to either a stereo or a compound microscope, to record their morphology and behaviour.

Laboratory feeding experiment

After starvation, medusae of both *Clytia* and *Obelia* were randomly divided into two groups per species, consisting of about 30 individuals each. In the feeding experiment,

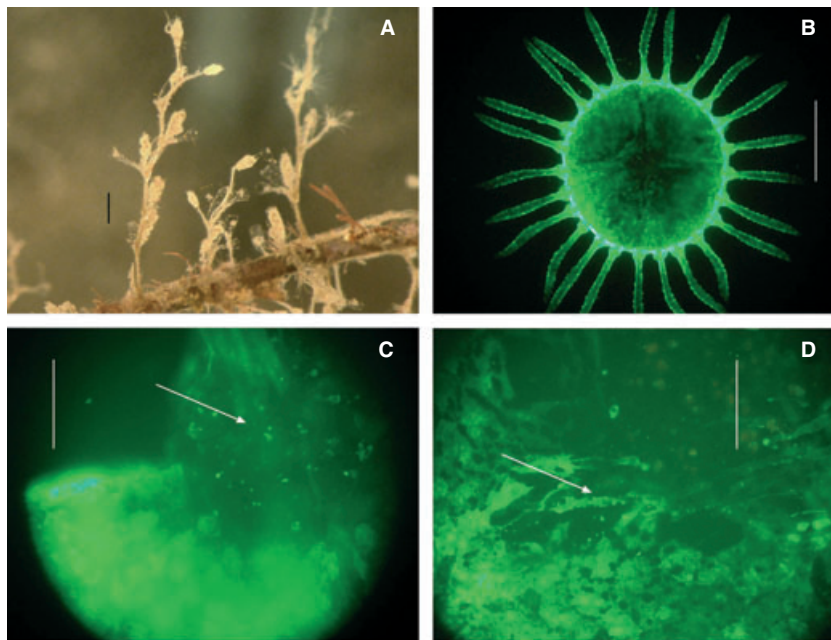


Fig. 1. A: Colony of *Obelia dichotoma* observed using light microscopy; B: *O. dichotoma* control medusa observed under an epifluorescence microscope (note the natural fluorescence of the medusa), the white ring indicates the stomach, the white square a radial canal; C: *O. dichotoma* treatment medusa observed after 30 min under an epifluorescence microscope (the arrow indicates the bacteria inside the stomach); D: *O. dichotoma* treatment medusa observed after 4 h under an epifluorescence microscope (note presence of bacteria inside a radial canal). Scales A: 0.5 mm; B: 0.25 mm; C–D: 50 μ m.

the first group (control) was placed in three small vessels (10 medusae per vessel) with sterile seawater (0.2 μ m pore-filtered). The controls allowed the condition of the medusae to be evaluated over the experimental period. The second group (treatment) was placed in three small vessels (10 medusae per vessel) with sterile seawater (0.2 μ m pore-filtered) plus genetically modified *Escherichia coli* expressing GFP at a final concentration of 10^6 cells ml^{-1} . Medusae were exposed to experimental conditions for a 4-h period. Every 30 min, medusae were washed with sterile seawater and observed under an epifluorescence microscope (Leitz-Leica Orthoplan) equipped with a halogen lamp (Hg 100). Blue light excitation with a BP 485/20 excitation filter, an FT 510 chromatic beam splitter and an LP 520 barrier filter were used to observe the slides prepared from each sample.

Plasmid construction and bacterial transformation

We cloned an *XbaI-HindIII* 900 bp fragment encoding GFP (Green Fluorescent Protein) Mut3 (Cormack *et al.* 1996) into the pNDE vector (Salvatore *et al.* 2000), obtaining pNDE-GFP. Subsequently, we introduced a 320 bp *BamHI-XbaI* fragment containing the promoter region of PorB gene (Wolff & Stern 1991) into pNDE-GFP, obtaining the pNDE-GFP-1 plasmid. This plasmid allows strong expression of GFP in *Escherichia coli*.

To obtain fluorescent bacteria, the *E. coli* genetically modified strain DH5 was transformed with the pNDE-GFP-1 construct as previously described (Inoue *et al.* 1990). Colonies obtained after transformation were resus-

pended with Marine Broth at the appropriate concentration to be added to the medusae.

Results

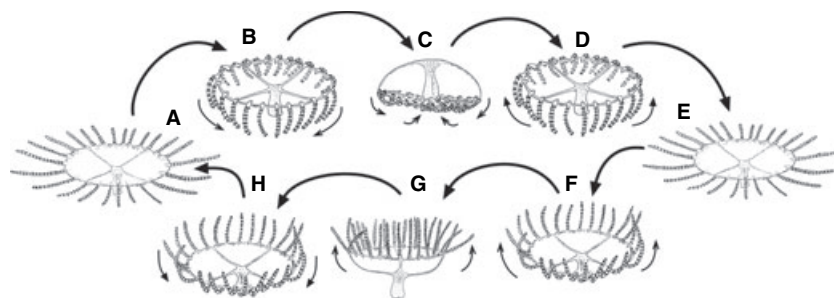
Feeding behaviour

Obelia dichotoma medusae, as observed for *O. longissima* by Boero & Sarà (1987), continuously pulsate, creating water currents towards the mouth. Under laboratory conditions, medusae usually perform this behaviour while positioned immediately below the water surface, with the manubrium hanging downwards. In this position, the pulsations of the umbrella do not produce movement, as the exumbrella is pressed against the air–water interface. The sequence of positions during the pulsation movements is as follows: a flat posture (Fig. 2A) is followed by a stage where the umbrella becomes concave and the tentacles push water (and food particles) towards the mouth (Fig. 2B and C); then the umbrella goes back to the initial (flat) condition (Fig. 2D and E); the medusa gains momentum to repeat the feeding stroke (Fig. 2F and G); finally, the umbrella returns to a flat posture (Fig. 2H).

Ingestion of genetically modified *Escherichia coli*

Both treated and untreated *Obelia* medusae are fluorescent. Untreated medusae show only their endogenous fluorescence (Fig. 1B), whereas treated medusae, fed with fluorescent and genetically modified *Escherichia coli*, show bacterial fluorescence in their gastric system (Fig. 1C).

Fig. 2. Swimming behaviour of *Obelia* medusae. A: flat posture; B–C: the umbrella becomes concave and the tentacles push water (and food particles) towards the mouth; D–E: the umbrella returns to a flat posture; F–G: the medusa gains momentum to repeat the feeding stroke; H: umbrella returns to a flat posture.



All *Obelia* medusae ingested an average of 20 bacteria after 30 min of treatment. The bacteria were concentrated in the stomach. Four hours from the beginning of the experiment, bacteria were concentrated in the radial canals of treated medusae (Fig. 1D).

Clytia hemisphaerica medusae did not ingest any bacteria during the 4-h study period, and swam up and down the water column in the rearing vessels.

Discussion

Our results show that newly released *Obelia* medusae perform a behaviour that leads to bacterial ingestion. Umbrellar pulsations are not used for movement but, instead, to create water currents that bring bacteria towards the mouth. Bacteria are then brought into the gastric cavity and circulate into the canal system.

Costello & Colin (2002) showed that some species of Hydromedusae can feed on ciliates (see also Parsons & Lalli 2003), but bacterial feeding by a specific behavioural pattern was still unreported. Newly released *O. dichotoma* medusae are active filter feeders that can concentrate bacteria in their gastric system. Contrary to *Obelia*, *C. hemisphaerica* medusae do not ingest bacteria. This probably depends on the different architecture of *Clytia* species, leading to a different trophic strategy: they swim to catch the prey, and their tentacles capture and bring it to the subumbrellar margin by a fast retraction (mouth to tentacle) (Miglietta *et al.* 2000).

Kubota (1981, 1999) showed that newly released *Obelia dichotoma* medusae from Japan can easily feed on crustaceans under laboratory conditions, whereas F. Boero and C. Gravili (unpublished data) were unsuccessful in feeding a similar diet to *O. dichotoma* from the Mediterranean. This discrepancy suggests that Japanese and Mediterranean *Obelia* populations may represent distinct species. In fact, some of Kubota's reared medusae already had immature gonads at liberation, whereas those from the Mediterranean Sea did not. The dissimilarity, however, might be due to intraspecific differences in the degree of medusa growth within the gonothea. Force-fed

Mediterranean medusae ingested crustacean nauplii but died thereafter, whereas Japanese medusae, referred to the same nominal species, reached maturity with this diet. The species-level taxonomy of *Obelia* is far from resolved (Cornelius 1999). Cornelius (1975, 1982) lumped 120 nominal *Obelia* species into three species (*O. bidentata*, *O. dichotoma*, *O. geniculata*). Later (Cornelius 1990), he recognised also *Obelia longissima* as valid.

Based on its particular medusa architecture, *Obelia* can be interpreted as a relatively recent evolution of the hydromedusan body plan, which is more linked to predation on Metazoa, towards a microphagous diet. Cnidaria are the most ancient Metazoa and usually feed on other Metazoa; logically, early in their evolutionary history, the organisms that make up their present diet (mainly crustaceans and fish eggs and larvae) were not available yet (Parsons 1979). In fact, there are cnidarian species with possibly 'ancestral' diets based on unicellular organisms like ciliates (Costello & Colin 2002). *Obelia*'s microphagous feeding habits could represent an example of the ancestral condition in Cnidaria, the other being the association with photosynthetic symbionts, as observed in many Anthozoa (Muller-Parker & Davy 2001; Secord 2001) and Hydrozoa medusae such as *Velilla* (Hollande & Carré 1974; Banaszak *et al.* 1993) and *Wuvula* (Bouillon *et al.* 1988). Because of its original body plan, however, *Obelia* is certainly not an ancestral medusa: it has many apomorphic characters that are thought to indicate saltational evolution from an ancestor with similar features to those of other campanulariids (Boero & Sarà 1987; Boero *et al.* 1996; Govindarajan *et al.* 2006). New characters in *Obelia*, lacking in other medusae even of the same Campanulariidae family, may reflect the accumulation of a considerable genetic variance along-side the evolutionary trend of the family, represented mainly by hydroids, which is abruptly expressed in the medusa. Furthermore, similarities between certain *Obelia* medusae features and hydranth features indicate that hydranth developmental pathways are involved in *Obelia* medusa development (Govindarajan *et al.* 2006).

Conclusion

The bacteriophagous diet of newly released *Obelia dichotoma* medusae, and the behavioural pattern involved in bacterial filtration described here, probably evolved secondarily from carnivorous ancestors. These features are thus analogous, and not homologous, to the ancestral feeding habits of hydrozoan medusae. Microphagy, however, might be more widespread in the Hydromedusae than usually thought, especially in the first stages of development, when the newly released medusae are too small to feed on the same food items as juvenile and adult stages.

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